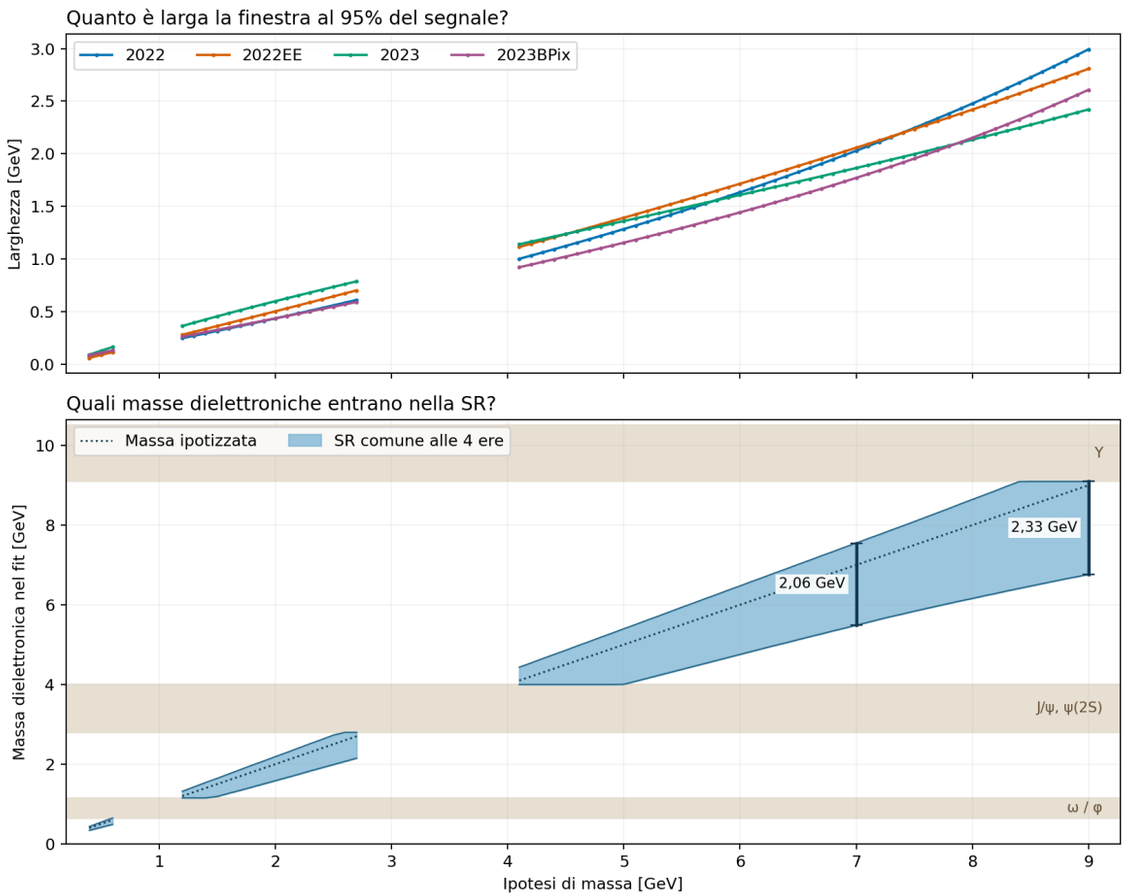
1 Choosing a common signal window

The scan combines 2022, 2022EE, 2023 and 2023BPix. For each mass and era, we find the shortest interval containing 95% of the full parametric double-sided Crystal Ball (DCB) signal. We take the lowest of the four lower edges and the highest of the four upper edges. This common interval covers at least 95% in every era.

The 261106 data use 5 MeV bins. We round the common window outwards to bin edges, then remove bins belonging to any control region (CR). The remaining bins define the signal region (SR). Its signal fraction can therefore fall below 95%. We test 69 masses from 0.4 to 9.0 GeV on a 0.1 GeV grid, skipping hypotheses centred in a CR.



At 0.6 GeV the common interval is 0.49313-0.65505 GeV. Rounding gives 0.490-0.660 GeV, and removing the light-resonance CR leaves 0.490-0.650 GeV (32 bins).

2 Control samples and resonance models

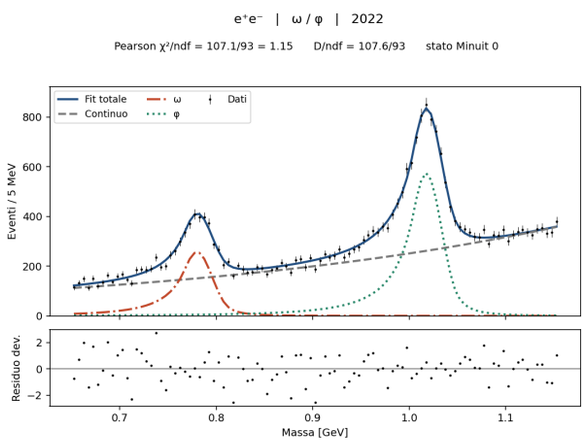
Three local windows group neighbouring SM peaks and include continuum on their flanks. The ranges below are the boundaries used on the 5 MeV grid. Each era is fitted independently in the CR prefit, with one signal-derived DCB per resonance, initialised at its nominal mass. Both peak yields, means and widths float in the prefit.

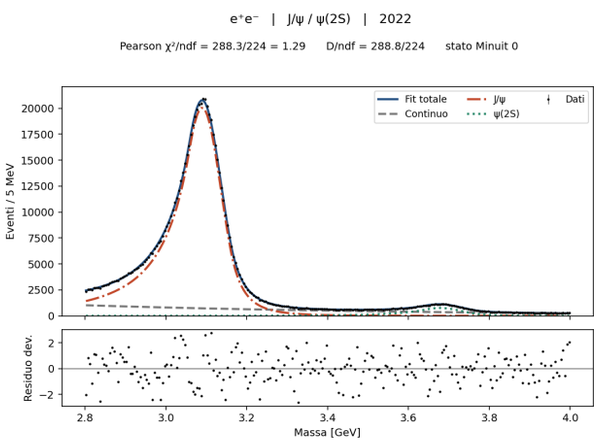
0.650-1.155 GeV | Omega and phi. Exponential continuum. Fix the DCB tails to their signal parametrisation.

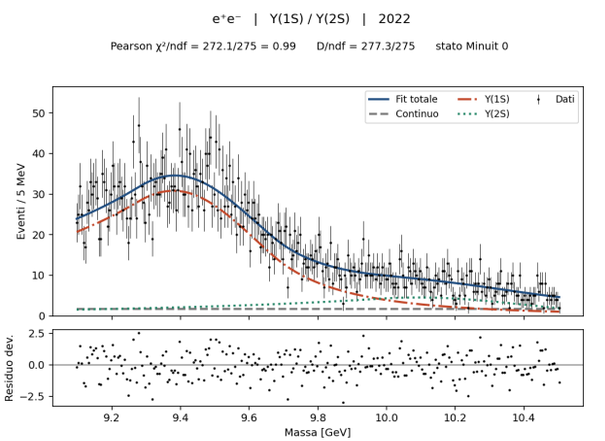
2.800-4.000 GeV | J/psi and psi(2S). A decreasing exponential gives a monotonic, convex continuum. Allow each left and right alpha and n to vary by up to 50% around its signal value in the prefit.

9.095-10.505 GeV | Upsilon(1S) and Upsilon(2S). Constant continuum and fixed signal-derived tails.

The light and Upsilon windows admitted simple continua in the earlier model comparisons. Charmonium retained structured residuals even after freeing its tails. We adopted that shape provisionally. The 2022 prefits below illustrate the fitted components and residuals for all three windows.







3 Simultaneous likelihood for each mass

For each candidate CR, integrate its fitted SM resonance tails over the actual SR bins and sum across its resonances and eras. Rank the CRs by predicted resonance yield. Use the smallest set of one or two CRs covering at least 95% of that yield, as recorded in the mass table. Continuum events do not enter this ranking.

Each era contributes one SR channel and one channel for each selected CR. Their binned Poisson likelihoods use expected counts integrated over the original bins. At 0.5 GeV the selected light and charmonium CRs give 12 channels in total. At 2.0 GeV charmonium alone gives eight. Each observed bin enters once per era.

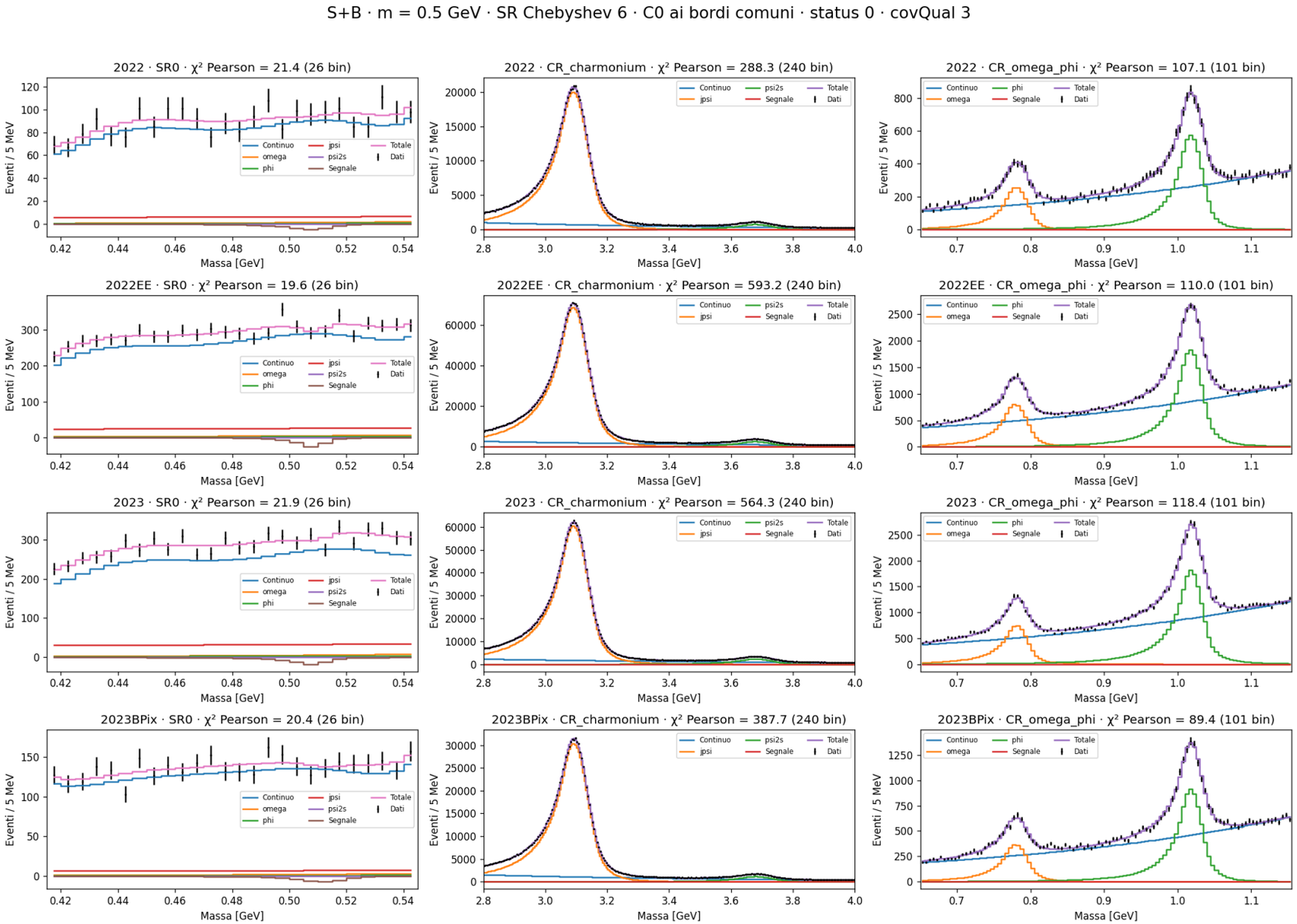
A single signal strength r scales the signal in every selected channel and era. The workspace imports the signal shape, yield and ten constrained nuisance parameters from the original signal model: electron ID, reconstruction and trigger, two luminosity terms, electron energy scale and four era-specific resolution terms. Shared nuisance constraints enter once. The original regional signal-rate convention is retained; its absolute normalisation still needs a provenance check.

Within each era, each SM peak has a common mean and normalisation across its CR and SR tail. The SM means and CR resonance and continuum yields float. Widths, tails and CR continuum shape parameters are fixed at their adopted prefit values. Their prefit uncertainties are not propagated. The SR continuum shape is profiled.

The baseline limit scan uses a positive degree-six Chebyshev continuum with value matching of the nonresonant density at shared edges. The later envelope tests use value matching of the total physical event density at actual SR/CR contacts. Neither fit matches across gaps. The slope is free. This distinction matters when interpreting the baseline limits alongside the bias tests.

Baseline Combine settings: AsymptoticLimits, Significance and FitDiagnostics use minimizer strategy 0 and MINIMIZER_freezeDisassociatedParams. FitDiagnostics additionally uses robustFit 1, saves shapes and normalisations, keeps failures and allows r from -30 to +30. Toy fits use their own configuration.

Simultaneous fit at 0.5 GeV

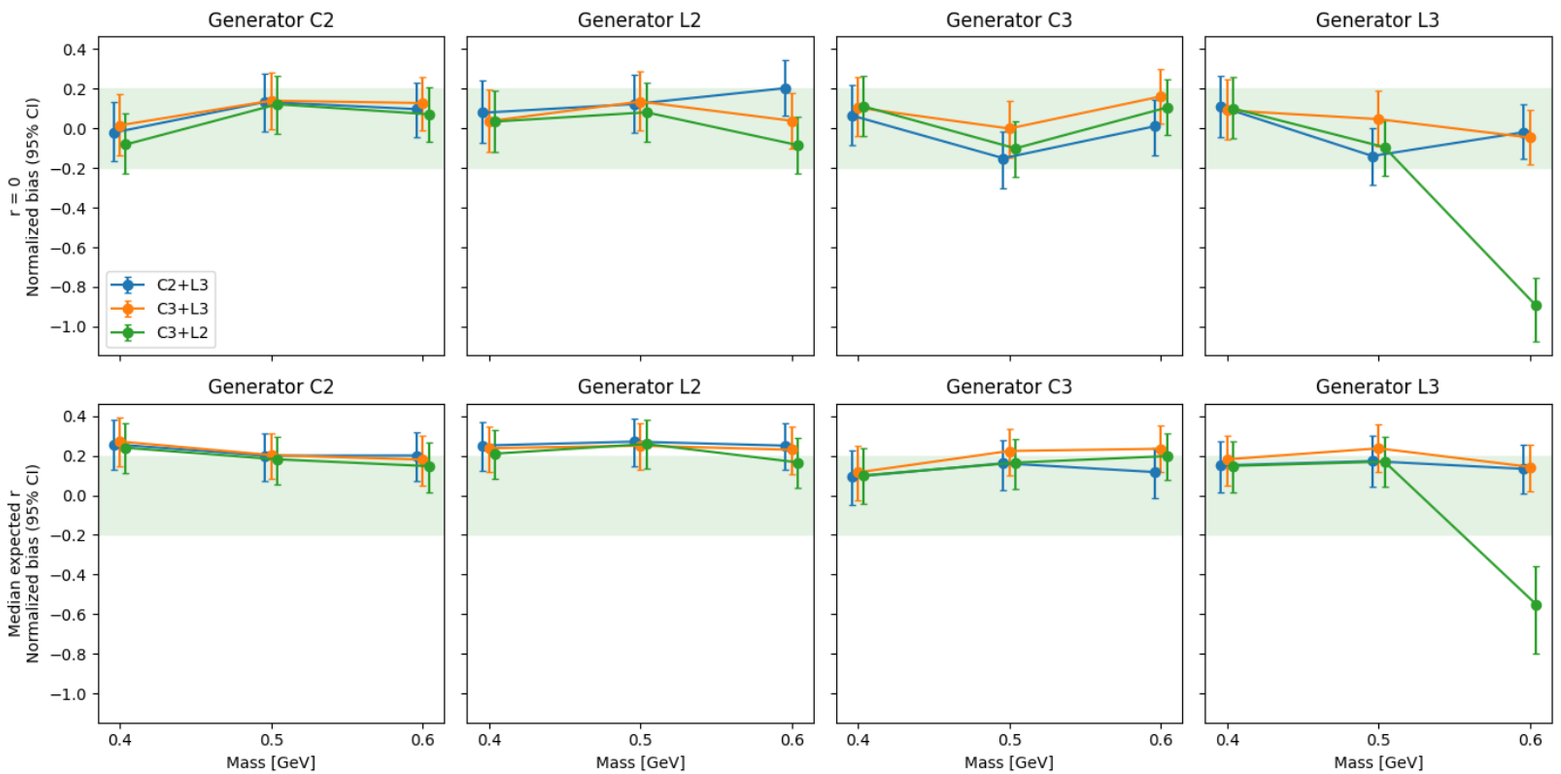


Baseline scan diagnostic from the fit gallery. The four eras use the same effective SR, 0.415-0.545 GeV. The light-resonance and charmonium control samples constrain the SM contributions in the joint likelihood.

4 Choosing the envelope below the phi

We generated 200 toys per mass (0.4, 0.5, 0.6 GeV), background generator (Chebyshev or log-polynomial of degree 2 or 3) and injection. The two injections were zero and the mass-specific median expected limit (11.25, 12.234375 and 14.25 respectively). This gives 4,800 distinct toys in 24 configurations.

We fit each toy with each candidate branch. For an envelope to enter the comparison both branches must have valid fits. The winner minimises the fitted deviance plus the parameter penalty (+8 for degree 2, +12 for degree 3). The bias is the mean of fitted minus injected r divided by its sample standard deviation. The bars show pointwise 95% bootstrap intervals. The plot compares identical toys valid for all three envelopes (166-200 per configuration).



Recommendation for 0.4-0.6 GeV: C2+L3 or C3+L3. Their maximum absolute biases are 0.271 and 0.271, with 7/24 and 8/24 point estimates outside +/-0.2. Neither meets that threshold everywhere. C3+L2 reaches -0.894 (95% interval -1.073 to -0.753) at 0.6 GeV for L3 generation with zero injection.

5 Choosing the envelope between phi and J/psi

The 16 mass hypotheses span 1.2 to 2.7 GeV in 0.1 GeV steps. At each mass we use the common four-era SR window and the CRs selected in the sliding-window table. The simultaneous fit includes signal in every selected channel and matches the total event density at shared SR/CR edges.

Chebyshev (C) and log-polynomial (L) fits of degrees 2 and 3 at 2.0 GeV provided an initial check of the SR residuals. We then tested whether the envelope could recover a known signal across the mass range. For each degree d, we generated 200 toys per mass with Cd and 200 with Ld. We fitted both branches to each toy and chose the lower penalized deviance among toys with two valid fits. The two branches of each envelope have the same penalty.

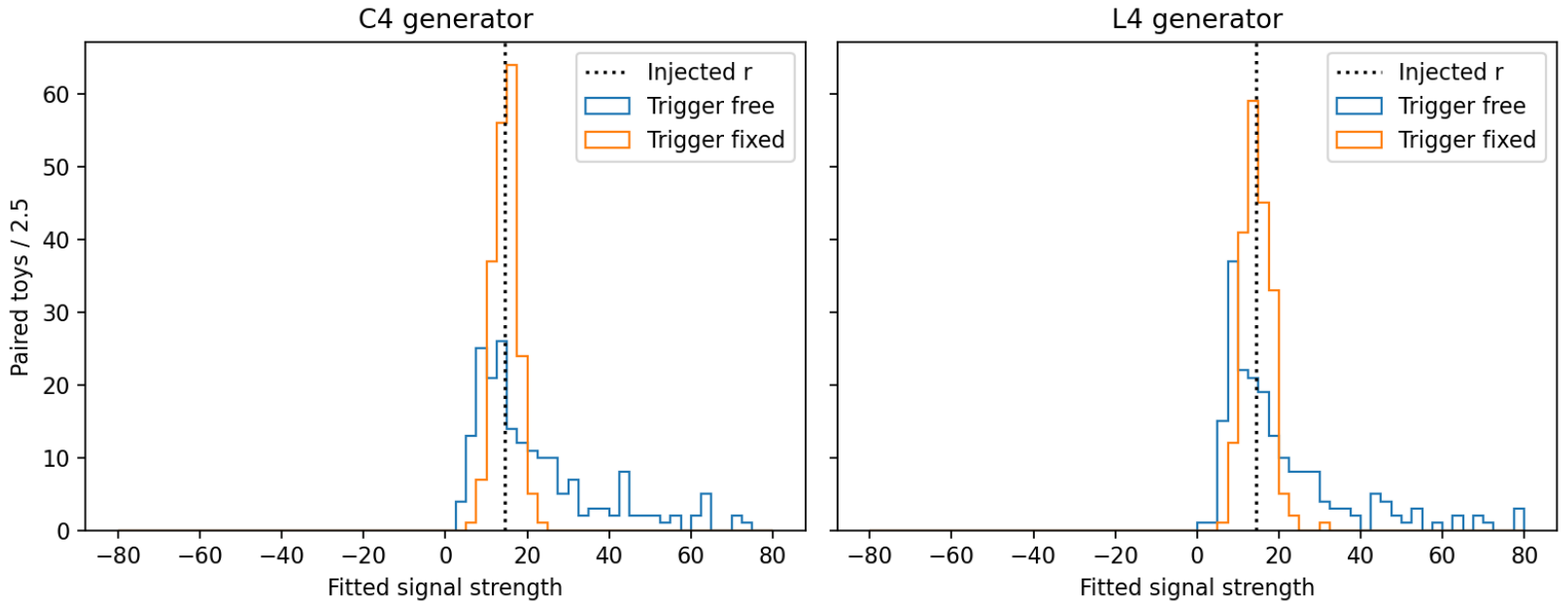
The normalized bias is the mean of fitted minus injected r divided by the sample standard deviation of fitted r. The table counts mass-generator combinations whose point estimate lies outside the target interval [-0.2, +0.2]. The positive injection equals the median expected limit at each mass.

Envelope	Injection	Valid / generated	Outside target	Maximum |bias|
C3 + L3	0	6,352 / 6,400	0 / 32	0.170
C3 + L3	Median expected	6,293 / 6,400	29 / 32	0.490
C4 + L4	0	6,359 / 6,400	0 / 32	0.176
C4 + L4	Median expected	6,277 / 6,400	28 / 32	0.489
C5 + L5	Median expected	5,142 / 5,200	25 / 26	0.494

The cubic and quartic envelopes meet the point-estimate target throughout the background-only tests. With signal injected, both exceed it in most configurations. The quintic test stopped after 13 masses (1.2, 1.8 and 2.7 GeV omitted). At 1.4 GeV its biases are +0.494 and +0.404 for C5 and L5 generation, close to the quartic results in Section 5.1.

5.1 Signal injection at 1.4 GeV and the trigger efficiency

At the median expected signal strength, the quartic envelope overestimates the injected signal on average. We fitted each toy twice with the same simultaneous SR + CR model and injected r = 14.390625. Only CMS_eff_trigger_e changed between fits. It was profiled in the nominal fit and fixed at its injected value of zero in the diagnostic fit. The histograms show the fitted signal strengths for the paired toys.



Normalized bias = mean(fitted r - injected r) / sample SD of fitted r. Each row uses the same seeds in both fits.

Generator	Valid pairs	Profiled	Fixed	Change [95% paired bootstrap CI]
Chebyshev 4	195 / 200	+0.489	+0.116	-0.373 [-0.528, -0.222]
Log-polynomial 4	199 / 200	+0.406	+0.071	-0.335 [-0.477, -0.196]

Fixing the trigger reduced the 90th percentile of fitted r from 44.7 to 18.3 (Chebyshev) and from 44.4 to 18.6 (log-polynomial). Six toys failed the two-branch validity requirement in the profiled fit. None of the selected estimates in the paired sample reached a fit boundary. The intervals use 10,000 paired resamples.

The measured trigger uncertainty remains in the nominal likelihood. Coverage requires a separate test of how often the upper limit includes the injected signal. This study measured bias at one mass and injection.

Injected-signal bias and validation of the quartic envelope

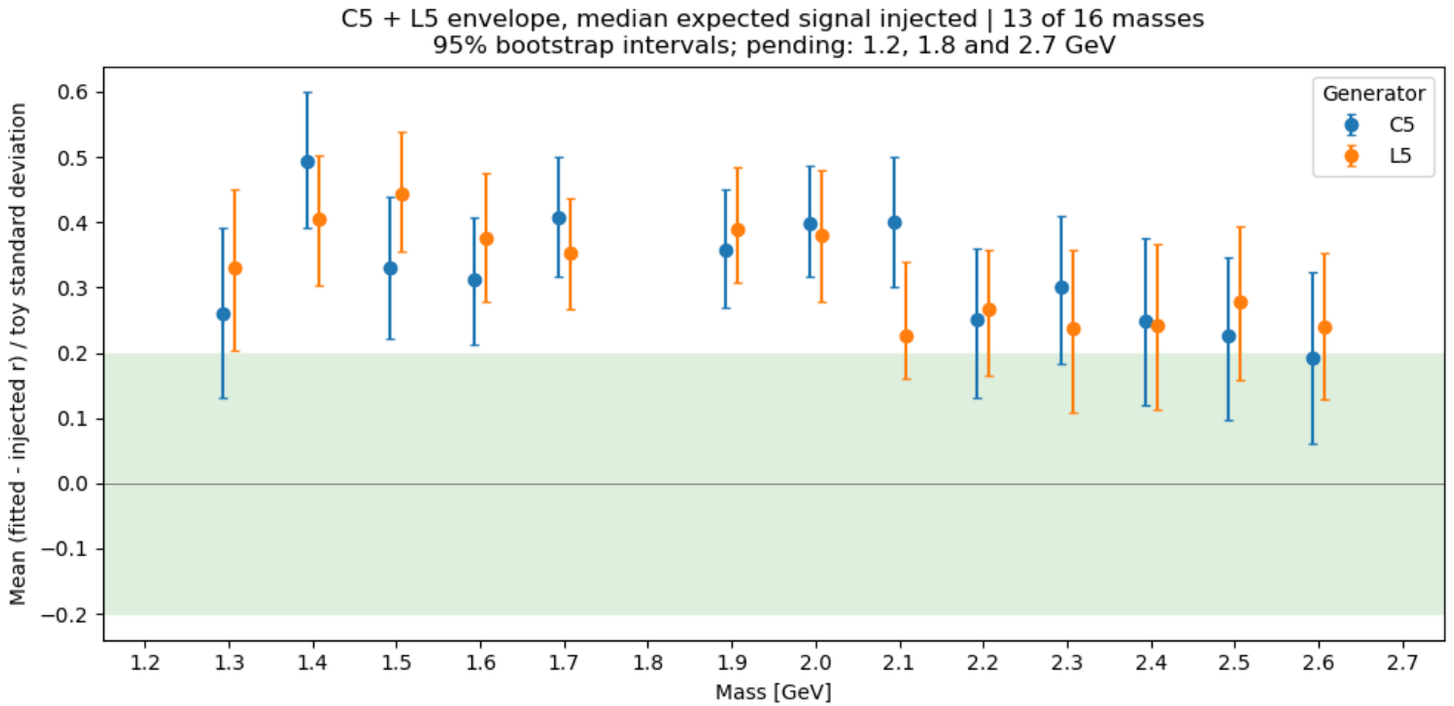


Figure. Quintic-envelope bias for the 13 completed masses with signal injected. For 14 of the 26 mass-generator combinations the entire pointwise bootstrap 95% interval is above +0.2.

Coverage at 1.4 GeV. We generate 200 toys per quartic generator at r = 14.390625 and profile the measured trigger-efficiency nuisance in the SR + CR fit. For each toy, Combine evaluates asymptotic CLs at the injected r. A value below 0.05 excludes the true injection. We report the covered fraction and its binomial interval for each generator, together with failed or numerically unreliable evaluations.

Fixed-trigger bias across all 16 masses. We refit the same median-injection toys with only CMS_eff_trigger_e fixed to its injected nominal value (zero). We compare the normalized bias to the profiled fit on common valid seeds, retaining both branch scores and the rejected-toy counts. The 400 toys at 1.4 GeV are reused. This tests whether the reduction seen there extends across the region.

Envelope choice remains open pending these checks. The nominal model retains the measured trigger uncertainty. The coverage and full-range fixed-trigger results are pending validation in this edition.